

E. Hidden Knowledge of the Ancients

time limit per test: 2 seconds
memory limit per test: 256 megabytes

In the world of *Deepwoken*, there exists an ancient artifact — *Tablet of Infinite Knowledge*, on which a sequence of n mysterious symbols (each symbol is an integer) is engraved.

It is said that the true power of the artifact can only be revealed by finding all *sacred fragments* — continuous segments of the tablet that contain exactly k distinct numbers, and their length must be between l and r (inclusive).

Formally: Given a sequence a of length n and integers k, l, r . You need to find the number of such boundaries b and c such that:

- $1 \leq b \leq c \leq n$;
- among the elements $[a_b, a_{b+1}, \dots, a_c]$ there are exactly k distinct numbers;
- $l \leq c - b + 1 \leq r$.

Input

Each test consists of several test cases.

The first line contains one integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The following describes the test cases.

The first line of each test case contains four integers: n, k, l , and r ($1 \leq k \leq n \leq 2 \cdot 10^5, 1 \leq l \leq r \leq n$).

The second line contains n numbers a_i ($1 \leq a_i \leq 10^9$) — the mysterious symbols.

It is guaranteed that the total value of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test, output a single integer on a separate line — the number of continuous subarrays that meet the specified conditions.

Example

input	Copy
5 1 1 1 1 5 5 2 2 3 1 2 1 3 2 6 3 1 6 1 2 3 1 2 3 4 1 1 2 7 7 7 7 7 3 2 4 1 2 1 2 3 2 1	
output	Copy
1 5 10 7 5	

Note

In the first test case $a = [5]$, there is only one subarray $[5]$, which has a length of 1 and contains exactly 1 distinct number.

In the fourth test case $a = [7, 7, 7, 7]$, any subarray of identical numbers gives exactly 1 distinct number. The start and end of possible subarrays are:

- Length 1: $[1, 1], [2, 2], [3, 3], [4, 4]$.
- Length 2: $[1, 2], [2, 3], [3, 4]$.

In total: 7.

In the fifth test case $a = [1, 2, 1, 2, 3, 2, 1]$:

- Length 2: all subarrays have only 2 distinct numbers.
- Length 3: $[3, 5], [5, 7]$.
- Length 4: $[2, 5], [3, 6], [4, 7]$.

In total: 5.

Codeforces Round 1054 (Div. 3)

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
342368167	Oct/07/2025 03:25	Happy New Year!

→ Problem tags

data structures

two pointers

*1500

No tag edit access

→ Contest materials

Announcement

Tutorial (en)